

Documentation

Useful information on fuels, tank systems and filtration

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Power. Passion. Partnership.

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Subject to alterations and amendments.

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1 Preface

1.1 Meaning of the fuels

The operational reliability of the complete engine plant, i.e. from the tank to the exhaust aftertreatment, is influenced strongly, among other things, by the fuel properties or by the fuel quality. Unsuitable fuel usually leads to a reduction in the engine service life and, in general, to engine damage.

The requirements of suitable fuels for MTU engines, including lists of fuels approved by MTU, are contained in the MTU Fluids and Lubricants Specifications (→ Page 5). The fuel properties, however, depend on many factors, e.g. region, time of year and storage. Measures must therefore be introduced to improve the fuel quality, or to keep it at a constant level.

This publication demonstrates the influence of the different fuel properties on the engine and describes how the fuel quality can be improved. Observance of the information and recommendations regarding the fuel used as well as the operating conditions increases the reliability of the engine and guarantees the operational data with the TBO.

If there are any problems with regard to observance of the fuel specifications, MTU can provide recommendations for improvement measures upon request.

MTU also assists in the selection and adaptation of suitable components for fuel-carrying components (e.g. prefilter). For this purpose, for example, samples can be taken from the fuel supply chain and examined and evaluated by MTU. This then forms the basis for any need for action.

Safety instructions for handling fuels and fuel systems



Fuels are hazardous substances. Certain regulations must be observed when handling, storing and disposing of these substances.

These regulations are contained in the manufacturers' instructions, legal requirements and technical guidelines valid in the individual countries. Great differences can apply from country to country and a generally valid guide to applicable regulations for fluids and lubricants is therefore not possible within this publication.

Users of the products named in these specifications are therefore obliged to inform themselves of the locally valid regulations. MTU Friedrichshafen GmbH accepts no liability whatsoever for improper or illegal use of the fluids and lubricants which it has approved.

1.2 Information on the use and applicability of this publication

Used symbols and means of representation

The following instructions are highlighted in the text and must be observed:



This symbol refers to:

- special material properties that have to be observed,
- instructions and activities that have to be observed, to prevent endangering personnel and damaging or destroying material.

The abbreviations used can be look up in the list of abbreviations (in Appendix A).

Applicability of the publication

This publication contains information on diesel fuels and kerosenes for MTU engines. The publication is independent of series and version.

The applicability and binding instructions of the respective engine documentation and the current MTU Fluids and Lubricants Specifications thus remain unaffected.

Overview of the Fluids and Lubricants Specifications

Publication number	Publication
A001061/..	MTU Fluids and Lubricants Specifications, general (except for series 1600 and 1800); commercial applications
AL01061/..	MTU Fluids and Lubricants Specifications for military land vehicles
AM01061/..	MTU Fluids and Lubricants Specifications for military marine applications
A001062/..	MTU Fluids and Lubricants Specifications for series 1800 "Power Pack"
A001063/..	MTU Fluids and Lubricants Specifications for series 1600
A001064/..	MTU Fluids and Lubricants Specifications for series 1600, 2000 and 4000 for engine-generator sets

2 Fuel properties and their effects on the engine

2.1 Total contamination and particle distribution

The total contamination, which is also specified in the fuel standard DIN EN 590, represents the total volume of all undissolved impurities (e.g. rust, dirt, dust, oxidation and bacterial degradation product) that can be separated by a diaphragm filter with a pore size of 0.8 µm. The total contamination is a binding upper limit for the quality of the fuel upon delivery.



Depending on the size and concentration, particles in the fuel can lead to a significantly reduced service life of filters and, in particular, injection components, and thus to considerable engine damage.

Particles can also enter the fuel system due to inattentiveness during assembly. It is therefore necessary to take special care with regard to contamination particularly during assembly in the field.

MTU has determined that the specification of total contamination alone is insufficient to guarantee reliable operation of modern diesel engines. Recommendations for refueling, care of the tank and prefiltration have therefore been worked out for reducing the risk of fuel contamination by water or particles, see Tank design (→ Page 30) and filter and water separator (→ Page 33). These specifications essentially guarantee that the permissible particle size distribution in the fuel is observed at the engine inlet. Under these conditions at the engine inlet, the filtration system works reliably and the engine can observe the promised service life.

Contamination classes

Characterization of the particle size distribution at MTU is based on the standard ISO 4406(c). A total of 29 contamination classes are defined for the three particle size ranges ($\geq 4 \mu\text{m}$, $\geq 6 \mu\text{m}$ and $\geq 14 \mu\text{m}$). The following table shows extracts.

Number of particles / ml	Contamination class
1.3 million to 2.5 million	28
...	...
5000 to 10000	20
2500 to 5000	19
1300 to 2500	18
640 to 1300	17
320 to 640	16
160 to 320	15
80 to 160	14
40 to 80	13
20 to 40	12
...	...
0.005 to 0.01	0

Table 1: Particle sizes and contamination classes

Particle size distribution for injection components

For injection components, suppliers of injection systems demand a particle distribution in the fuel before the HP pump of 12/9/6 according to ISO 4406(c). This corresponds to:

Number of particles / ml	Particle size
20 to 40	$\geq 4 \mu\text{m}$
2.5 to 5	$\geq 6 \mu\text{m}$
0.32 to 0.64	$\geq 14 \mu\text{m}$

Table 2: Particle size distribution 12/9/6 for injection components

Together with the filter manufacturers, MTU has developed and evaluated systems that make it possible to meet these specifications. Only the observance of these specifications guarantees long, safe operation of the engine.

Particle size distribution in the tank

MTU defines a maximum permissible particle size distribution of 18/17/14 in the tank before the fuel filtration chain in accordance with the definition (→ Table 1) in order to achieve, with suitable filtration, a particle distribution of 12/9/6 before the HP pump. This corresponds to:

Number of particles / ml	Particle size
1300 to 2500	$\geq 4 \mu\text{m}$
640 to 1300	$\geq 6 \mu\text{m}$
80 to 160	$\geq 14 \mu\text{m}$

Table 3: Particle size distribution 18/17/14 in the tank

This criterion is usually met upon delivery from refinery. A large proportion of the contamination is thus removed in the fuel logistics chain before reaching the engine. A suitable refueling system, tank design, tank care and prefilter system can guarantee that the engine reaches the required service life and TBO, see Tank design (→ Page 30) and filter and water separator (→ Page 33).

If tank quality 18/17/14 is not guaranteed, additional filtration components are required.

2.2 Density

The density of fuel varies according to the chemical composition of the fuel. If the density deviates significantly from the specified values, it can be assumed that there are also significant deviations in other fuel parameters, which can then result in engine operation problems.

Minor deviations in density do not usually cause engine damage. The influence on combustion can therefore be regarded as minimal.



Also note the density in conjunction with the calorific value (→ Page 18) because this is used to define the volumetric energy content of the fuel. Deviations can result in serious engine damage.

2.3 Viscosity

Viscosity is a unit for measuring the thickness of the fuel. The kinematic viscosity of fuels is typically measured at a temperature of 40 °C. The viscosity is generally highly dependent on the temperature.

The viscosity influences the flow behavior of the fuel. It initially influences pressure losses or volume flows under given pressure conditions. Microscopically, a changed viscosity can interfere with local flow patterns and result in damage due to cavitation.

MTU engines are designed to deal with a wide range of fuel viscosities. Information on the permissible viscosity range is available in the Fluids and Lubricants Specifications. MTU has no validated findings related to operation outside this range. As the above-described damage cannot be excluded with the use of fuels with changed viscosity, the engine must not be operated in such cases without further examinations.

Through restriction of the permissible temperature range (e.g. through specification of maximum temperatures for "arctic fuel" or minimum temperatures for more viscous fuels), it is easily possible to extend the range of permissible fuels with regard to standard viscosity at 40 °C. These fuels then receive limited approval from MTU with the specification of permissible temperatures.

2.4 Flashpoint and boiling curve

The flashpoint is essential a safety-related parameter. The flashpoint is used to divide mineral oil products into hazard classes. The flashpoint is the temperature at which sufficient fuel evaporates in an open vessel at room temperature to create a flammable mixture that can be externally ignited. Engine operation is not influenced by the flashpoint.

Diesel fuel is a mixture of hydrocarbons, which boil approximately in a range between 170 °C and 380 °C. According to DIN EN590, three characteristic points are specified for the boiling curve of diesel fuel: 250 °C, 350 °C and the temperature at which 95% of the fuel is evaporated.



Lower boiling fuel tends to initiate cavitation damage in fuel systems, which can be coupled quickly with considerable engine damage.

If the boiling point is too high, the droplet size in the injection jet increases, which can lead to increased pressure gradients in the combustion chamber and increased soot formation. This seriously impairs the operation of components of the exhaust gas recirculation and diesel particulate filtration systems.

MTU therefore only permits operation with fuels outside the specifications following an intensive examination.